

# List 2

## Questions 1, 5, 10 and 13

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### Question 1

The induced norm is  $\|u\|_V = \sqrt{(u, u)_V}$ . Note that, if  $v, w \in V$ :

$$\begin{aligned}
 (v, w)_V &= \\
 &= (v, w - v + v)_V = \\
 &= (v, w - v)_V + (v, v)_V = \\
 &= (v, w - v)_V + \|v\|_V^2 = \\
 &= (v - w + w, w - v)_V + \|v\|_V^2 = \\
 &= (v - w, w - v)_V + (w, w - v)_V + \|v\|_V^2 = \\
 &= -\|v - w\|_V^2 + \|w\|_V^2 + \|v\|_V^2 - (w, v)_V = \\
 &= -\|v - w\|_V^2 + \|w\|_V^2 + \|v\|_V^2 - (v, w)_V
 \end{aligned}$$

Isolating  $(v, w)_V$  gives us:

$$(v, w)_V = \frac{1}{2} \left( \|v\|_V^2 - \|v - w\|_V^2 + \|w\|_V^2 \right) \quad (1)$$

Note that  $(\|v\|_V - \|w\|_V)^2 = \|v\|_V^2 - 2\|v\|_V \|w\|_V + \|w\|_V^2$ . Substituting in equation 1 gives us:

$$(v, w)_V = \frac{1}{2} \left( (\|v\|_V - \|w\|_V)^2 + 2\|v\|_V \|w\|_V - \|v - w\|_V^2 \right)$$

Taking the absolute value in both sides:

$$\begin{aligned}
 |(v, w)_V| &= \\
 &= \frac{1}{2} \left| (\|v\|_V - \|w\|_V)^2 - \|v - w\|_V^2 + 2\|v\|_V \|w\|_V \right| \leq \\
 &\leq \frac{1}{2} \left| (\|v\|_V - \|w\|_V)^2 - \|v - w\|_V^2 \right| + \|v\|_V \|w\|_V \leq \\
 &\leq \|v\|_V \|w\|_V
 \end{aligned}$$

Therefore  $|(v, w)_V| \leq \|v\|_V \|w\|_V$ .

## Question 5

For  $\vec{\alpha} = (2, 0, 0)$  we have:

$$D^{\vec{\alpha}} f = \frac{\partial^2 f}{\partial x_1^2}$$

For  $\vec{\alpha} = (0, 1, 1)$ :

$$D^{\vec{\alpha}} f = \frac{\partial^2 f}{\partial x_2 \partial x_3}$$

## Question 10

a)

First of all

$$\frac{1}{r} + \frac{1}{r'} = \frac{1}{r} + \frac{r-1}{r} = \frac{r}{r} = 1$$

So we can apply Hölder's inequality to these factors.  
Note that:

$$\| |f|^p \|_{L^1(\Omega)} = \int_{\Omega} |f|^p dx = \|f\|_{L^p(\Omega)}^p \quad (2)$$

and

$$\begin{aligned} \| |f|^p \|_{L^1(\Omega)} &= \\ &= \| |f|^p \cdot 1 \|_{L^1(\Omega)} \leq \\ &\leq \| |f|^p \|_{L^r(\Omega)} \|1\|_{L^{r'}(\Omega)} = \\ &= \left( \int_{\Omega} |f|^{p \cdot \frac{r}{r'}} dx \right)^{\frac{r}{r'}} \left( \int_{\Omega} 1^{r'} dx \right)^{\frac{1}{r'}} = \\ &= |\Omega|^{\frac{1}{r'}} \left( \int_{\Omega} |f|^q dx \right)^{\frac{p}{q}} = \\ &= |\Omega|^{\frac{q-p}{q}} \left( \int_{\Omega} |f|^q dx \right)^{\frac{p}{q}} \end{aligned}$$

Using equation 2:

$$\|f\|_{L^p(\Omega)} = \|f\|_{L^1(\Omega)}^{\frac{1}{p}} \leq |\Omega|^{\frac{q-p}{qp}} \|f\|_{L^q(\Omega)} = |\Omega|^{\frac{1}{p} - \frac{1}{q}} \|f\|_{L^q(\Omega)}$$

If  $\Omega$  is not bounded, then  $|\Omega|$  is not finite, which gives us nothing useful, as  $\|f\|_{L^p(\Omega)}$  could be infinite.

The demonstration above shows that if  $f \in L^q(\Omega)$  and  $\Omega$  is bounded, then  $\|f\|_{L^p(\Omega)} < \infty \implies f \in L^p(\Omega)$

b)

First we're finding  $s$  such that  $f \in L^1(\Omega) \iff \int_{\Omega} |f(x)| dx < \infty$ .

$$\int_{\Omega} |f(x)| dx = \int_0^1 x^{-s} dx$$

- If  $s < 1$ ,

$$\int_{\Omega} |f(x)| dx = \left. \frac{x^{1-s}}{1-s} \right|_0^1 = \frac{1^{1-s}}{1-s} - \frac{0}{1-s} = \frac{1}{1-s} < \infty$$

- If  $s > 1$

$$\int_{\Omega} |f(x)| dx = \left. \frac{1}{(1-s)x^{s-1}} \right|_0^1 = \frac{1}{1-s} - \lim_{x \rightarrow 0^+} \frac{1}{(1-s)x^{s-1}} = \infty$$

- If  $s = 1$

$$\int_{\Omega} |f(x)| dx = \ln(x) \Big|_0^1 = \infty$$

Therefore,  $f \in L^1(\Omega)$  when  $s < 1$ .

Now we find  $s$  such that  $f \in L^2(\Omega) \iff \int_{\Omega} |f(x)|^2 dx < \infty$ .

$$\int_{\Omega} |f(x)|^2 dx = \int_0^1 x^{-2s} dx = \int_{\Omega} x^{-t} dx, \text{ with } t = 2s$$

We know this converges for  $t < 1 \implies s < \frac{1}{2}$ .

So,  $\forall s \in [\frac{1}{2}, 1)$ ,  $f \in L^1(\Omega)$  but  $f \notin L^2(\Omega)$ .

This shows the inclusion is strict.

c)

$$\int_{\Omega} |f(x)|^2 dx = \int_1^{\infty} \frac{1}{x^2} dx = \left. \frac{-1}{x} \right|_1^{\infty} = - \lim_{x \rightarrow \infty} \frac{1}{x} - \left( \frac{-1}{1} \right) = 0 + 1 < \infty$$

So  $f \in L^2(\Omega)$ .

$$\int_{\Omega} |f(x)| dx = \int_1^{\infty} \frac{1}{x} dx = \ln(x) \Big|_1^{\infty} = \lim_{x \rightarrow \infty} \ln(x) = \infty$$

So  $f \notin L^1(\Omega)$ .

This shows the boundedness of  $\Omega$  is necessary for the inclusion to be valid.

### Question 13

a)

Let  $\varphi \in C_0^\infty(\Omega)$ .

$$\int_{-1}^1 g(x)\varphi'(x)dx = \int_{-1}^0 \varphi'(x)dx - \int_0^1 \varphi'(x)dx = \varphi(x)|_{-1}^0 - \varphi|_0^1 = 2\varphi(0)$$

Supposing  $h$  is a weak derivative for  $g$ , we have:

$$\begin{aligned} - \int_{-1}^1 h(x)\varphi(x)dx &= \int_{-1}^1 g(x)\varphi'(x)dx = 2\varphi(0) \implies \\ \implies \int_{-1}^1 h(x)\varphi(x)dx &= -2\varphi(0) = -2 \int_{-1}^1 \delta_0\varphi(x)dx \end{aligned}$$

As  $g$  is differentiable for  $x \neq 0$ , we must have  $h = g'$  for  $x \neq 0$ . But then  $h(x) \neq 0$  only at  $x = 0$ . But  $\{0\}$  is a set with measure zero. If  $h(0)$  is finite, then we'd have

$$\int_{-1}^1 h(x)\varphi(x)dx = 0, \forall \varphi \in C_0^\infty(\Omega)$$

which is an absurd. So  $h$  must be a Dirac's delta.

b)

First we show that  $\text{supp}(\varphi_n) \subset (\frac{-1}{n}, \frac{1}{n})$ :

Take  $x_0 \in \text{supp}(\varphi_n)$ .  $\forall \varepsilon > 0$ ,  $\exists x_1 \in B_\varepsilon(x_0)$  such that  $\varphi_n(x_1) \neq 0$ . So,  $\varphi(nx_1) \neq 0 \implies nx_1 \in (-1, 1) \implies x_1 \in (\frac{-1}{n}, \frac{1}{n})$ .

Take  $\varepsilon = \frac{1}{2} \min\{|x_0 - \frac{1}{n}|, |x_0 + \frac{1}{n}|\}$

Suppose by absurdity  $x_0 \geq \frac{1}{n}$

$$|x_0 - x_1| \geq |x_0| - |x_1| \geq \frac{1}{n}$$

Therefore

$$\begin{aligned} |x_1| &\geq \\ &\geq \frac{1}{n} - |x_0 - x_1| \geq \\ &\geq \frac{1}{n} - \frac{1}{2} \min\left\{\left|x_0 - \frac{1}{n}\right|, \left|x_0 + \frac{1}{n}\right|\right\} = \frac{1}{n} - \frac{1}{2} \left(x_0 - \frac{1}{n}\right) = \\ &= \frac{3}{2n} - \frac{x_0}{2} \geq \frac{3}{2n} - \frac{1}{2n} = \frac{1}{n} \end{aligned}$$

Absurd, because  $|x_1| < \frac{1}{n}$ . Analogous for  $x_0 \leq \frac{-1}{n}$ .

$\varphi_n$  is continuous, as  $\varphi_1$  is continuous and  $\text{supp}(\varphi_n) \subset (\frac{-1}{n}, \frac{1}{n}) \subset (-1, 1)$ . So  $\varphi_n \in \mathcal{D}(-1, 1)$

As consequence of this fact,

$$\int_{-1}^1 h\varphi_n dx = -2\varphi_n(0) = -2$$

But,  $0 \leq \varphi_n \leq 1$ , which gives

$$\left| \int_{-1}^1 h\varphi_n dx \right| = \left| \int_{-\frac{1}{n}}^{\frac{1}{n}} h\varphi_n dx \right| \leq \int_{-\frac{1}{n}}^{\frac{1}{n}} |h|\varphi_n dx \leq \int_{-\frac{1}{n}}^{\frac{1}{n}} |h| dx \xrightarrow{n \rightarrow \infty} 0$$

Using the absolute continuity of the Lebesgue integral to justify the convergence. Absurd, because

$$\left| \int_{-1}^1 h\varphi_n dx \right| = |-2\varphi_n(0)| = 2$$

Therefore,  $g$  does not have a weak derivative.